Practical 2d: Interfaces

AIM : To demonstrate the use of interfaces in C# — defining contracts that classes must implement.

PROGRAM CODE

```
using System;
namespace InterfaceDemo
{
   // Define interfaces
   interface IShape
   {
      double Area();
      double Perimeter();
      void Display();
   }

   interface IPrintable
   {
      void Print();
   }

   class Circle : IShape, IPrintable
   {
      private double radius;
      public Circle(double r) { radius = r; }

      public double Area() => Math.PI * radius * radius;
      public double Perimeter() => 2 * Math.PI * radius;
      public void Display()
      {
         Console.WriteLine($"Circle: Radius={radius}");
         Console.WriteLine($"   Area      = {Area():F2}");
         Console.WriteLine($"   Perimeter = {Perimeter():F2}");
      }
      public void Print() => Console.WriteLine("[Printable] Circle Info Printed.");
   }

   class Rectangle : IShape
```

```
    {
        private double length, width;
        public Rectangle(double l, double w) { length = l; width = w; }

        public double Area() => length * width;
        public double Perimeter() => 2 * (length + width);
        public void Display()
        {
            Console.WriteLine($"Rectangle: Length={length}, Width={width}");
            Console.WriteLine($"   Area      = {Area():F2}");
            Console.WriteLine($"   Perimeter = {Perimeter():F2}");
        }
    }
    class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("=== Interface Demo ===");
            IShape c = new Circle(7.0);
            c.Display();
            ((IPrintable)c).Print();

            Console.WriteLine();
            IShape r = new Rectangle(5.0, 3.0);
            r.Display();
            Console.ReadKey();
        }
    }
}
```

OUTPUT

```
=== Interface Demo ===
Circle: Radius=7
  Area      = 153.94
  Perimeter = 43.98
[Printable] Circle Info Printed.

Rectangle: Length=5, Width=3
  Area      = 15.00
  Perimeter = 16.00
```